Relazione sull'attività svolta dal 1 Novembre 2022 al 28 Febbraio 2025

Emanuele Di Marco

L'attività da me svolta in questo periodo si inserisce nella ricerca di fenomeni non previsti dal Modello Standard delle particelle elementari nell'ambito dei dei fenomeni elettrodeboli e del bosone di Higgs, nonché di misure di parametri fondamentali dello stesso, soprattutto nell'ambito dell'esperimento *CMS* al *Large Hadron Collider* (LHC) del *CERN* di Ginevra. Allo stesso tempo ho contribuito alle misure relative all'esperimento *CYGNO*, per la ricerca di segnali di materia oscura, utilizzando tutti i dati presi dal primo prototipo, *LIME*, il quale è stato installato, commissionato e ha preso dati in maniera continuativa nei *Laboratori Nazionali del Gran Sasso* (LNGS).

In questo periodo, nell'ambito dell'attività dell'esperimento CMS, mi sono dedicato a studi sul bosone di Higgs. Dalla primavera del 2022, infatti, l'acceleratore LHC ha fornito collisioni protone-protone (pp) ad un centro di massa maggiore di quello della presa dati del Run-2: $\sqrt{s} = 13.6\,\mathrm{TeV}$, e l'esperimento CMS ha raccolto circa $41.5\,\mathrm{fb}^{-1}$, $32.7\,\mathrm{fb}^{-1}$ e $122.2\,\mathrm{fb}^{-1}$ nel 2022, 2023 e 2024, rispettivamente. Quindi da un lato è stato possibile misurare le sezioni d'urto di produzione del bosone di Higgs a questa energia del centro di massa mai raggiunta prima, e confrontarle con le predizioni teoriche. Dall'altro lato, con questa luminosità integrata, confrontabile a quella raccolta con i precedenti run di LHC, è stato possibile fare nuove misure rese accessibili dal campione di dati due volte maggiore, e dalla maggiore sezione d'urto di produzione a $13.6\,\mathrm{TeV}$. Sono uno degli autori principali, nonché supervisor di due studenti di dottorato di cui sono relatore, di due analisi attualmente in fase di pubblicazione.

La prima è la misura della larghezza intrinseca del bosone di Higgs, che essendo prevista dal Modello Standard essere $\Gamma_H = 4.1\,\mathrm{MeV}$, è di circa tre ordini di grandezza inferiore rispetto alla risoluzione sperimentale dei canali con alta risoluzione di massa, $H \to ZZ \to 4\ell$ e $H \to \gamma\gamma$. Questa è una misura innovativa

che sfrutta lo shift di massa indotto dall'interferenza quantistica tra l'ampiezza risonante e quella del processo non risonante $pp \to \gamma\gamma$. La sensibilità è quella di una decina di MeV, un ordine di grandezza maggiore al limite superiore ottenuto dagli eventi *off-shell* di $H \to 4\ell$, ma con assunzioni teoriche molto minori e comunque un ordine di grandezza migliore del vincolo dalla misura diretta. La seconda analisi dei dati di CMS di cui sono autore principale insieme a una studentessa di dottorato di cui sono relatore, sfrutta anch'essa il decadimento $H \to \gamma\gamma$, e ricerca possibili accoppiamenti anomali del bosone di Higgs ai bosoni vettori ($V = W, Z$), tramite la produzione VBF e associata VH.

Dal 2021 fino a Settembre 2023 sono stato il chair *Publication Committee* del gruppo del bosone di Higgs di CMS. Nel corso di questo periodo ho coordinato le pubblicazioni di tutti gli articoli (circa 30) sul bosone di Higgs. Dal 1 Settembre 2023 ad oggi sono il convener dell'intero gruppo di fisica dell'Higgs, coordinando il lavoro di circa 300 persone coinvolte nelle relative analisi dei dati. Durante questo periodo ho condotto ai risultati preliminari, o alla pubblicazione su rivista, circa 50 analisi. La mia attività in questo ambito va dalla pianificazione dei risultati di fisica delle linee di ricerca sull'Higgs sia nell'immediato futuro, che con il dataset completo di LHC prima dell'upgrade alla fase 2 (Run2+Run3, previsto terminare la presa dati a metà del 2026). Essa include la supervisione dei sottogruppi ($H \to \gamma\gamma$, $H \to ZZ$, $H \to WW$, $H \to b\bar{b}$, *HLepRare*), nel quale l'aspetto fondamentale è quello della guida e revisione di tutte le analisi, dall'impostazione iniziale della strategia, alla comprensione dei risultati, fino alla fase di pubblicazione su rivista. Una parte considerevole dell'impegno è quello del coordinamento tra il gruppo di analisi e quelli di *offline and computing* e delle performance dei rivelatori strettamente legate ai risultati di fisica.

I risultati in cui ho dato un contributo personale determinante, e pubblicate nel periodo considerato, sono: la combinazione ATLAS+CMS che ha permesso la prima evidenza del decadimento raro $H \to Z\gamma$ [1], i vincoli sull'autoaccoppiamento del bosone di Higgs dalla combinazione delle misure di singolo H e doppio H [2], la prima osservazione del decadimento $H \to b\bar{b}$ con la produzione VH [3], la prima misura della produzione rara bbH, attraverso i decadimenti $H \to \tau\tau$ o $H \to WW$ [4], la prima misura della produzione $ttH(b\bar{b})$, includendo anche stati finali completamente adronici (sottomessa a rivista, in fase di pubblicazione). Lo sviluppo di algoritmi di *tagging* basati su Machine Learning, come *Graph Neural Networks*, atti a determinare il sapore dei quark originanti i jet adronici, ha permesso di determinare il limite superiore più basso per l'accoppiamento di Yukawa del bosone di Higgs ai quark di seconda generazione, $H \to c\bar{c}$, attraverso la produzione associata VH [5]. L'applicazione

di queste tecniche, applicate agli stati finali con alto boost di Lorentz del bosone di Higgs, ha reso accessibili alcuni decadimenti completamente adronici, che fino ad oggi erano al di là della portata sperimentale, permettendo di abbattere di ordini di grandezza il fondo di multijet da QCD. Esempi su cui il mio contributo nella definizione della strategia di analisi e revisione in qualità di convener dell'Higgs ha permesso la pubblicazione in questo periodo sono: $H \to \tau\tau$ [6] e la produzione in gluon-fusion e VBF di $H \to b\bar{b}$ [7], in un regime in cui i prodotti di decadimento adronici (dei τ o dei quark b) sono inglobati in uno stesso jet ricostruito a causa dell'alto boost. L'analisi della sotto-struttura e delle correlazioni dei prodotti dell'adronizzazione sono risultati fondamentali, ma ha richiesto un lavoro raffinato nella modellizzazione dell'adronizzazione con la simulazione, e inoltre calibrazioni precise della risposta di questi algoritmi tra dati e simulazioni. Queste tecniche rendono possibile ampliare la sensibilità a meccanismi rari di produzione del bosone di Higgs, e in particolare quello del doppio-H non risonante, che offre il metodo migliore per vincolare l'accoppiamento $H - H$, il quale determina la forma del potenziale del campo di Higgs, che è correlato alla stabilità quantistica dell'Universo. Insieme a un terzo dottorando, di cui sono relatore, mi sto cominciando ad occupare di $HH \to b\bar{b}\tau\tau$, uno dei 3 canali che, grazie alla combinazione di alto *Branching Ratio* e relativamente alto rapporto segnale/fondo, sarà fondamentale per determinare i limiti più stringenti con i dati della fase-1 di LHC. Personalmente sto iniziando l'analisi dei dati di Run3 di LHC, ma ho coordinato come convener del gruppo di Higgs l'analisi dei dati di Run2 [8].

Inoltre, sebbene non ancora pubblicate, ma con risultati preliminari già presentati a conferenze internazionali, ho partecipato e coordinato le prime misure della sezione d'urto, fiduciale e differenziale, del bosone di Higgs a $\sqrt{s} = 13.6\,\mathrm{TeV}$, nei canali $H \to \gamma\gamma$ e $H \to ZZ \to 4\ell$, nonché molteplici altre misure di precisione che hanno permesso di vincolare effetti di nuova fisica a scale di energia inaccessibili direttamente a LHC attraverso le interpretazioni nell'ambito delle *Teorie di Campo Effettive*.

In questo periodo è stata completata, resa pubblica e sottomessa su rivista "Nature", la misura della massa del bosone W, $m_W = 80\,360.2(99)\,\mathrm{MeV}$, in accordo con la predizione del Modello Standard, e con la maggior precisione sperimentale mai raggiunta. Questo risultato completa l'analisi dei dati di CMS del 2020 che era stata usata per la misura delle sezioni d'urto differenziali, in stati di elicità e di carica, su cui ho lavorato come autore principale, negli anni precedenti al periodo in considerazione, con la stessa tecnica di estrazione del segnale usata per la misura di m_W. Essa infatti ha permesso di vincolare le sistematiche delle PDF a LHC che erano uno dei fattori limitanti per m_W a un

collisore pp. L'aggiunta della calibrazione di scala di momento dei muoni con una precisione di $\approx 10^{-4}$ ha permesso quindi finalmente l'estrazione di m_W con la precisione di $10\,\mathrm{MeV}$. Esso è il primo passo verso una misura che raggiunga la precisione teorica del fit elettrodebole di circa $5\,\mathrm{MeV}$, che quindi potrebbe permettere un test della consistenza del Modello Standard, e in particolare delle correzioni quantistiche della massa del W: per raggiungerlo servirà analizzare tutti i dati di Run2 con la stessa precisione, e riducendo ancora di più le assunzioni teoriche sul modello di produzione del bosone W a LHC.

Durante questo periodo, ho continuato ad essere co-responsabile del centro di calcolo (*Tier-2*) per i dati di CMS di Roma, un sistema condiviso con l'esperimento ATLAS. Esso è costituito da dieci rack che contengono qualche centinaio di server di elaborazione dati (per un totale di circa 2500 core) e archiviazione dei dati (per un totale di circa 2 PB). Il centro di calcolo permette l'eleborazione di circa il 20% dei dati raccolti dal LHC ogni anno, ed è connesso al sistema di *grid* mondiale. In questi anni mi sono occupato della pianificazione degli upgrade e degli acquisti per il Tier-2 di CMS, per un investimento medio di circa 100 k€ all'anno, nonché dell'installazione delle risorse di calcolo e di storage dati acquistate con i fondi relativi al *PNRR*.

Nel periodo di interesse ho continuato la partecipazione allo sviluppo dell'esperimento CYGNO, un rivelatore per la ricerca diretta di materia oscura di tipo *Weak Interacting Massive Particle* (WIMP), che si manifesti nei rinculi delle ipotetiche particelle sui nuclei di atomi di gas, con energia cinetica di pochi keV. Nel periodo 2022-2025 il prototipo *LIME* è stato portato ai LNGS, e ho quindi partecipato sia al commissioning del rivelatore che alle calibrazioni in energia, con il primo dataset completo raccolto *underground*. Fino a Settembre 2023 sono stato il coordinatore del gruppo di analisi dati dell'esperimento CYGNO, e mi sono occupato del codice e dell'infrastruttura per la ricostruzione *prompt* dei dataset raccolti dall'esperimento, usata da tutti i membri della collaborazione internazionale. Ho sviluppato ulteriormente tali algoritmi, adattandoli all'*occupancy* media più ridotta rispetto ai test con il prototipo in superficie, soprattutto nella configurazione con uno *shield* di $5\,\mathrm{cm}$ di rame radiopuro intorno alla TPC. Sono l'autore principale delle misure descritte nel'articolo [9], le quali hanno permesso di individuare aspetti specifici, quali quello dell'elettroluminescenza, descritto nell'articolo [10].

Durante il periodo in considerazione mi sono occupato anche di didattica relativa al mio campo di interesse, come esercitatore per il corso di Fisica Nucleare e Subnucleare 1 nell'a.a. 2022-2023, come tutor di esperienze di laboratorio per il corso di Laboratorio di Fisica Nucleare 2 (dal 2019 ad oggi), come relatore di 3 tesi magistrali, 3 tesi di dottorato ancora in corso e numerose

dissertazioni triennali.

Inoltre in questo periodo sono stato referee articoli scientifici per la rivista *PLB* ("Physics Letters B", nel settore di "Particle Physics, Nuclear Physics and Cosmology"), per la rivista *JHEP* ("Journal of High Energy Physics"), per la rivista *ROPP* ("Reports on Progress in Physics"), per la rivista *MLST* ("Machine Learning: Science and Technology"), ed editor della sezione "Upgrades in High Energy Physics Experiments" della rivista *Symmetry* di MDPI.

10 pubblicazioni scelte dal 1 Novembre 2022 al 28 Febbraio 2025

[1] G. Aad et al. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC. *Physical Review Letters*, 132(2), January 2024.

[2] Aram Hayrapetyan et al. Constraints on the Higgs boson self-coupling from the combination of single and double Higgs boson production in proton-proton collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Phys. Lett. B*, 861:139210, 2025.

[3] Armen Tumasyan et al. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H \to b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Phys. Rev. D*, 109(9):092011, 2024.

[4] Aram Hayrapetyan et al. Search for bottom quark associated production of the standard model Higgs boson in final states with leptons in proton-proton collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Phys. Lett. B*, 860:139173, 2025.

[5] Armen Tumasyan et al. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Phys. Rev. Lett.*, 131(6):061801, 2023.

[6] Aram Hayrapetyan et al. Measurement of the production cross section of a Higgs boson with large transverse momentum in its decays to a pair of τ leptons in proton-proton collisions at s=13TeV. *Phys. Lett. B*, 857:138964, 2024.

[7] Aram Hayrapetyan et al. Measurement of boosted Higgs bosons produced via vector boson fusion or gluon fusion in the H $\to$b$\bar{\text{b}}$ decay mode using LHC proton-proton collision data at $\sqrt{s} = 13\,\mathrm{TeV}$. *JHEP*, 12:035, 2024.

[8] Armen Tumasyan et al. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Phys. Lett. B*, 842:137531, 2023.

[9] Fernando Domingues Amaro et al. A $50\,\mathrm{l}$ Cygno prototype overground characterization. *Eur. Phys. J. C*, 83(10):946, 2023.

[10] F. D. Amaro et al. Secondary scintillation yield from GEM electron avalanches in He-CF_4 and He-CF_4-isobutane for CYGNO — Directional Dark Matter search with an optical TPC. *Phys. Lett. B*, 855:138759, 2024.

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Lista di tutte le pubblicazioni dal 1 Novembre 2022 al 28 Febbraio 2025

[1] Vladimir Chekhovsky et al. Search for medium effects using jet axis decorrelation in inclusive jets from PbPb collisions at $\sqrt{s_{\mathrm{NN}}}$ = 5.02 TeV. 2 2025.

[2] Vladimir Chekhovsky et al. Development of systematic uncertainty-aware neural network trainings for binned-likelihood analyses at the LHC. 2 2025.

[3] Vladimir Chekhovsky et al. Evidence for similar collectivity of high transverse momentum particles in pPb and PbPb collisions. 2 2025.

[4] Aram Hayrapetyan et al. Observation of Λ hyperon local polarization in pPb collisions at $\sqrt{s_{\mathrm{NN}}}$ = 8.16 TeV. 2 2025.

[5] Vladimir Chekhovsky et al. Search for bosons of an extended Higgs sector in b quark final states in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 2 2025.

[6] Vladimir Chekhovsky et al. Search for γH production and constraints on the Yukawa couplings of light quarks to the Higgs boson. 2 2025.

[7] Vladimir Chekhovsky et al. Measurements of Higgs boson production cross section in the four-lepton final state in proton-proton collisions at $\sqrt{s}$ = 13.6 TeV. 1 2025.

[8] Vladimir Chekhovsky et al. Search for a heavy pseudoscalar Higgs boson decaying to a 125 GeV Higgs boson and a Z boson in final states with two

tau and two light leptons in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 1 2025.

[9] Aram Hayrapetyan et al. Evidence for CP violation and measurement of CP-violating parameters in $\mathrm{B}_\mathrm{s}^0 \to \mathrm{J}/\psi\,\phi$(1020) decays in pp collisions at $\sqrt{s} =$ 13 TeV. 12 2024.

[10] Aram Hayrapetyan et al. Search for a neutral gauge boson with nonuniversal fermion couplings in vector boson fusion processes in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 12 2024.

[11] Vladimir Chekhovsky et al. Identification of low-momentum muons in the CMS detector using multivariate techniques in proton-proton collisions at $\sqrt{s}$ = 13.6 TeV. 12 2024.

[12] Vladimir Chekhovsky et al. Determination of the strong coupling and its running from measurements of inclusive jet production. 12 2024.

[13] Aram Hayrapetyan et al. Measurement of the W boson decay branching fraction ratio $\mathcal{B}(W \to cq)/\mathcal{B}(W \to \mathrm{q\bar{q}'})$ in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 12 2024.

[14] Aram Hayrapetyan et al. Measurement of light-by-light scattering and the Breit-Wheeler process, and search for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{\mathrm{NN}}}$ = 5.02 TeV. 12 2024.

[15] Vladimir Chekhovsky et al. High-precision measurement of the W boson mass with the CMS experiment at the LHC. 12 2024.

[16] Aram Hayrapetyan et al. Search for heavy neutral resonances decaying to tau lepton pairs in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 12 2024.

[17] Vladimir Chekhovsky et al. Model-agnostic search for dijet resonances with anomalous jet substructure in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 12 2024.

[18] Aram Hayrapetyan et al. Measurement of the inclusive WZ production cross section in pp collisions at $\sqrt{s}$ = 13.6 TeV. 12 2024.

[19] Aram Hayrapetyan et al. Search for heavy neutral Higgs bosons A and H in the $\mathrm{t\bar{t}Z}$ channel in proton-proton collisions at 13 TeV. 11 2024.

[20] Aram Hayrapetyan et al. Proton reconstruction with the TOTEM Roman pot detectors for high-β^* LHC data. 11 2024.

[21] Vladimir Chekhovsky et al. Constraints on standard model effective field theory for a Higgs boson produced in association with W or Z bosons in the H$\rightarrow \mathrm{b\bar{b}}$ decay channel in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 11 2024.

[22] Aram Hayrapetyan et al. Search for rare decays of the Z and Higgs bosons to a J/ψ or ψ(2S) meson and a photon in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 11 2024.

[23] Aram Hayrapetyan et al. Angular analysis of the $\mathrm{B}^0 \rightarrow \mathrm{K}^*(892)^0\mu^+\mu^-$ decay in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 11 2024.

[24] Aram Hayrapetyan et al. Reweighting simulated events using machine-learning techniques in the CMS experiment. 11 2024.

[25] Aram Hayrapetyan et al. Search for a heavy resonance decaying into a Z and a Higgs boson in events with an energetic jet and two electrons, two muons, or missing transverse momentum in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[26] Aram Hayrapetyan et al. Measurements of inclusive and differential cross sections for top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[27] Aram Hayrapetyan et al. Measurement of the inclusive $\mathrm{t\bar{t}}$ cross section in final states with at least one lepton and additional jets with 302 pb^{-1} of pp collisions at $\sqrt{s}$ = 5.0 TeV. 10 2024.

[28] Aram Hayrapetyan et al. Search for excited tau leptons in the $\tau\tau\gamma$ final state in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[29] Aram Hayrapetyan et al. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[30] Aram Hayrapetyan et al. Search for the Higgs boson decays to a ρ0, ϕ, or K∗0 meson and a photon in proton-proton collisions at s=13TeV. *Phys. Lett. B*, 862:139296, 2025.

[31] Aram Hayrapetyan et al. Search for high-mass resonances in a final state comprising a gluon and two hadronically decaying W bosons in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[32] A. Hayrapetyan et al. Performance of the CMS high-level trigger during LHC Run 2. *JINST*, 19(11):P11021, 2024.

[33] Aram Hayrapetyan et al. Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[34] Armen Tumasyan et al. Correlations between azimuthal anisotropy and mean transverse momentum in pp, pPb, and peripheral PbPb collisions. 10 2024.

[35] Aram Hayrapetyan et al. Study of same-sign W boson scattering and anomalous couplings in events with one tau lepton from pp collisions at $\sqrt{s}$ = 13 TeV. 10 2024.

[36] Aram Hayrapetyan et al. Energy-scaling behavior of intrinsic transverse momentum parameters in Drell-Yan simulation. 9 2024.

[37] Aram Hayrapetyan et al. Measurement of the Higgs boson mass and width using the four-lepton final state in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 9 2024.

[38] Aram Hayrapetyan et al. Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using lepton+jets events from proton-proton collisions at s=13 TeV. *Phys. Rev. D*, 110(11):112016, 2024.

[39] Aram Hayrapetyan et al. Search for light long-lived particles decaying to displaced jets in proton–proton collisions at $\sqrt{s} = 13.6$ TeV. *Rept. Prog. Phys.*, 88(3):037801, 2025.

[40] Aram Hayrapetyan et al. Bottom quark energy loss and hadronization with B^+ and B_s^0 nuclear modification factors using pp and \PbPb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV. 9 2024.

[41] Aram Hayrapetyan et al. Measurement of inclusive and differential cross sections of single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s}$ = 13.6 TeV. *JHEP*, 01:107, 2025.

[42] Aram Hayrapetyan et al. Pseudorapidity distributions of charged hadrons in lead-lead collisions at sNN=5.36TeV. *Phys. Lett. B*, 861:139279, 2025.

[43] Aram Hayrapetyan et al. Search for dark matter produced in association with a pair of bottom quarks in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 8 2024.

[44] Aram Hayrapetyan et al. Measurement of the Drell–Yan forward-backward asymmetry and of the effective leptonic weak mixing angle in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 8 2024.

[45] Aram Hayrapetyan et al. Measurement of the inclusive cross sections for W and Z boson production in proton-proton collisions at $\sqrt{s}$ = 5.02 and 13 TeV. 8 2024.

[46] Aram Hayrapetyan et al. Search for bottom quark associated production of the standard model Higgs boson in final states with leptons in proton-proton collisions at s=13TeV. *Phys. Lett. B*, 860:139173, 2025.

[47] Aram Hayrapetyan et al. Test of lepton flavor universality in semileptonic $\mathrm{B_c^+}$ meson decays in proton-proton collisions at $\sqrt{s}$ = 13 TeV. 8 2024.

[48] Aram Hayrapetyan et al. Model-independent search for pair production of new bosons decaying into muons in proton-proton collisions at $\sqrt{s}$ = 13 TeV. *JHEP*, 12:172, 2024.

[49] Aram Hayrapetyan et al. Search for flavor-changing neutral current interactions of the top quark mediated by a Higgs boson in proton-proton collisions at 13 TeV. 7 2024.

[50] Aram Hayrapetyan et al. Multiplicity dependence of charm baryon and baryon meson production in pPb collisions at $\sqrt{s_{\mathrm{NN}}}$ = 8.16 TeV. 7 2024.

[51] Aram Hayrapetyan et al. Constraints on the Higgs boson self-coupling from the combination of single and double Higgs boson production in proton-proton collisions at s=13TeV. *Phys. Lett. B*, 861:139210, 2025.

[52] Aram Hayrapetyan et al. Measurement of the $B_{\mathrm{s}}^0 \to J/\psi K_{\mathrm{S}}^0$ effective lifetime from proton-proton collisions at $\sqrt{s} = 13$ TeV. *JHEP*, 10:247, 2024.

[53] Armen Tumasyan et al. Search for resonant pair production of Higgs bosons in the $\mathrm{b\bar{b}b\bar{b}}$ final state using large-area jets in proton-proton collisions at $\sqrt{s}$ = 13 TeV. *JHEP*, 02:040, 2025.

[54] Aram Hayrapetyan et al. Search for long-lived heavy neutral leptons in proton-proton collision events with a lepton-jet pair associated with a secondary vertex at $\sqrt{s}$ = 13 TeV. *JHEP*, 02:036, 2025.

[55] Aram Hayrapetyan et al. Measurement of the $\mathrm{t\bar{t}H}$ and tH production rates in the H $\to \mathrm{b\bar{b}}$ decay channel using proton-proton collision data at $\sqrt{s}$ = 13 TeV. 7 2024.

[56] Aram Hayrapetyan et al. Measurement of boosted Higgs bosons produced via vector boson fusion or gluon fusion in the H $\rightarrow$b$\bar{\text{b}}$ decay mode using LHC proton-proton collision data at $\sqrt{s}$ = 13 TeV. *JHEP*, 12:035, 2024.

[57] Aram Hayrapetyan et al. Search for nuclear modifications of B^+ meson production in pPb collisions at $\sqrt{s_{\text{NN}}}$ = 8.16 TeV. 7 2024.

[58] Aram Hayrapetyan et al. Observation of double J/ψ meson production in pPb collisions at sNN=8.16 TeV. *Phys. Rev. D*, 110(9):092002, 2024.

[59] Aram Hayrapetyan et al. Measurement of the polarizations of prompt and non-prompt Image 1 and ψ(2S) mesons produced in pp collisions at s=13TeV. *Phys. Lett. B*, 858:139044, 2024.

[60] Aram Hayrapetyan et al. Search for a resonance decaying to a W boson and a photon in proton-proton collisions at $\sqrt{s}$ = 13 TeV using leptonic W boson decays. *JHEP*, 09:186, 2024.

[61] Fernando Domingues Amaro et al. Enhancing the light yield of He:CF_4 based gaseous detector. *Eur. Phys. J. C*, 84(10):1122, 2024.

[62] Aram Hayrapetyan et al. Measurement of inclusive and differential cross sections for W^+W^- production in proton-proton collisions at $\sqrt{s}$ = 13.6 TeV. *Phys. Lett. B*, 861:139231, 2025.

[63] Aram Hayrapetyan et al. Observation of quantum entanglement in top quark pair production in proton–proton collisions at $\sqrt{s} = 13$ TeV. *Rept. Prog. Phys.*, 87(11):117801, 2024.

[64] Aram Hayrapetyan et al. Observation of $\gamma\gamma \rightarrow \tau\tau$ in proton-proton collisions and limits on the anomalous electromagnetic moments of the τ lepton. *Rept. Prog. Phys.*, 87(10):107801, 2024.

[65] F. D. Amaro et al. Secondary scintillation yield from GEM electron avalanches in He-CF_4 and He-CF_4-isobutane for CYGNO — Directional Dark Matter search with an optical TPC. *Phys. Lett. B*, 855:138759, 2024.

[66] F. D. Amaro et al. Charge amplification in low pressure CF_4:SF_6:He mixtures with a multi-mesh ThGEM for directional dark matter searches. *JINST*, 19(06):P06021, 2024.

[67] Aram Hayrapetyan et al. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at s=13TeV. *Phys. Lett. B*, 860:139067, 2025.

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